# Multi-UAV Search and Coordination using Advanced Cell-DEVS Models

Bahareh Pourahmad, Rachid Fahmi, Maryam Bazyari
Department of Systems and Computer Engineering
Carleton University
Ottawa, Ontario, Canada

***Abstract*—This project presents an advanced Cell-DEVS model for multi-UAV search and coordination implemented in Cadmium. The work extends the single-UAV search model from Assignment 2 into a multi-agent spatial model that incorporates multiple UAVs, uncertainty-aware search, revisit penalties, and shared-information coordination. The environment is represented as a non-wrapped $30 \times 30$ grid in which each cell stores a target-belief value and additional search-related attributes.**

**We compare six base scenarios: a single-UAV baseline, independent multi-UAV search, partitioned deployment, shared-information coordination, and two diffusion-sensitivity experiments. In addition, three stress-scenario experiments are included as a robustness and scalability extension on a larger grid with more UAVs and hotspot locations. Results are evaluated using coverage, overlap, revisit intensity, first-detection time, and hotspots found.**

**The results show that local coordination strategies significantly influence search behavior. In particular, partitioned deployment reduces overlap, shared-information coordination alters exploration patterns but does not fully eliminate redundancy, and the diffusion parameter strongly affects how quickly the probability field spreads across the environment.**

**The project demonstrates that Cell-DEVS provides a suitable formal framework for modeling distributed autonomous search using local decision rules and reproducible simulation experiments. Overall, the results highlight the trade-off between exploration efficiency and redundancy and emphasize the importance of coordination design in decentralized multi-UAV systems.**

## I. Introduction

### A. Motivation

Autonomous UAV search has become increasingly important in surveillance, search-and-rescue, disaster-response assessment, and environmental monitoring. In many realistic situations, one UAV alone is too slow to cover the area of interest, making multi-UAV coordination necessary. However, the addition of more agents does not automatically improve system performance. Without appropriate coordination, multiple UAVs may repeatedly visit the same cells, follow similar gradients, and waste time exploring already attractive regions. Prior work on automated UAV search and cooperative multi-UAV search highlights the need to balance coverage, redundancy, and detection performance [3], [4].

This trade-off makes the problem especially suitable for modeling and simulation. A spatial search process evolves over time, depends on local interactions, and often involves emergent behavior that is difficult to analyze analytically. We therefore use the Cell-DEVS formalism, which provides a rigorous event-based framework for representing spatially distributed systems.

### B. Goals

The main goals of this project are:

- to build a multi-UAV search model using Cell-DEVS and Cadmium;
- to extend the Assignment 2 model with new rules and richer cell behavior;
- to compare different coordination strategies under a common simulation framework;
- to analyze how environment structure and diffusion influence search performance.

### C. Contributions

The project makes the following contributions:

- extends a single-UAV search model into a multi-agent Cell-DEVS framework with decentralized decision-making;
- introduces an enhanced cell state including uncertainty, revisit memory, and coordination penalties;
- develops a local scoring mechanism that balances exploration and exploitation using only neighborhood information;
- implements multiple coordination strategies, including independent search, partitioned initialization, and shared-information interaction;
- provides a fully reproducible simulation pipeline, including automated experiment execution, metric extraction, and visualization;
- analyzes the trade-off between exploration efficiency and redundancy using both qualitative and quantitative results.

Unlike many multi-UAV search approaches that rely on explicit communication or centralized coordination, this work shows that local coordination patterns can be induced through purely local interactions. In particular, the use of penalty-based mechanisms within the Cell-DEVS framework enables UAVs to balance exploration and redundancy without global knowledge or direct message passing.

This highlights the capability of Cell-DEVS to capture distributed behavior in systems where coordination patterns can arise from simple local rules. The experimental results confirm that even lightweight coordination mechanisms strongly influence search efficiency and spatial exploration patterns, revealing clear trade-offs between coverage, overlap, and detection performance.

## II. Background

The Discrete Event System Specification (DEVS) formalism provides a rigorous mathematical framework for modeling discrete-event systems [1]. A DEVS atomic model is defined as

$$M = \langle X, S, Y, \delta_{int}, \delta_{ext}, \lambda, ta \rangle$$

where:

- $X$ is the set of input events,
- $S$ is the set of states,
- $Y$ is the set of output events,
- $\delta_{int}$ is the internal transition function,
- $\delta_{ext}$ is the external transition function,
- $\lambda$ is the output function,
- $ta$ is the time advance function.

DEVS is attractive because it separates system behavior from the simulation mechanism. Models can be built modularly and then simulated in a rigorous, reusable way. For spatial systems, Cell-DEVS extends DEVS by associating state with cells in a grid and by defining local evolution based on neighborhood relationships [2]. This makes Cell-DEVS especially useful for diffusion, local propagation, occupancy, movement, and agent-based spatial interaction.

In autonomous UAV search, probability maps are frequently used to represent target likelihood or belief over space [3]. Meanwhile, cooperative multi-UAV search emphasizes the need for coordination policies that reduce overlap while maintaining broad coverage [4]. Our work combines these two ideas in a formal Cell-DEVS setting: a belief-like probability field evolves spatially, while multiple UAVs make local movement decisions influenced by probability, uncertainty, and coordination effects.

This project also matches the course emphasis on advanced Cell-DEVS models because it is not only a reimplementation of a prior assignment, but an enhanced version with new rules, new state variables, different experiment settings, and richer analysis.

## III. Models Defined

### A. System Description

The system represents a search mission over a finite $30 \times 30$ environment. The area is discretized into cells, and time advances in discrete Cell-DEVS steps. Each cell stores local information about target likelihood and search history. Multiple UAVs move through the environment, making decisions based only on local neighborhood information. Their goal is to detect hotspots while exploring efficiently and minimizing unnecessary redundancy.

The final model extends a simpler version by introducing:

- multiple UAV agents instead of a single agent,
- optional obstacle constraints that influence movement and diffusion,
- spatial variation in cell behavior (e.g., zone-based effects),
- local uncertainty to encourage exploration,
- visit and shared penalties to discourage repeated search and clustering.

### B. Environment Structure

The grid is non-wrapped, so border cells do not connect around the edges. The environment contains several spatial elements that influence both UAV motion and probability evolution:

- **hotspots**: cells with persistent target signal that maintain high probability values and act as target sources;
- **free space**: the remaining grid where UAVs can move and search normally;
- **obstacles**: blocked cells that constrain UAV movement and suppress probability propagation through those locations;
- **zones**: spatially differentiated regions in which local behavior may vary, for example through zone-dependent scoring or diffusion effects.

The initial positions of UAVs, hotspots, and other environment features are defined in the experiment configuration files. Figure 1 illustrates a representative environment used in the experiments. Although the baseline environment is visually simple, the underlying model supports richer spatial structure than open free space alone. This is important because the search process is affected not only by target locations, but also by local constraints and region-dependent behavior that shape movement and probability evolution.

### C. Cell State

Each cell stores a structured state:

- **prob**: belief or detection probability in $[0, 1]$,
- **uav**: UAV occupancy or movement code,
- **zone**: environment type,
- **uncertainty**: local search uncertainty,
- **visit_penalty**: discourages revisiting previously searched cells,
- **shared_penalty**: discourages clustering near other UAVs.

The UAV state code is interpreted as:

- 0: no UAV in the cell,
- 100: active UAV currently occupying the cell,
- 200: locked UAV, indicating target detection or hold state,
- 1 to 8: directional move code toward one of the Moore-neighborhood cells.

### D. Implementation-to-Model Mapping

To make the relationship between the conceptual model and the implementation explicit, Table I summarizes the meaning of the main state variables used in each Cell-DEVS cell.

This table clarifies that the model does not rely on a single scalar probability alone. Instead, each cell stores both environmental information and search-history information, allowing the system to represent local attraction, prior exploration, and decentralized coordination effects simultaneously.

TABLE I
MAPPING BETWEEN CELL-DEVS STATE VARIABLES, MODELING MEANING, AND IMPLEMENTATION ROLE.

| Variable | Modeling meaning | Implementation role |
|---|---|---|
| `prob` | target belief / diffusion field | updated via diffusion and UAV interaction |
| `uav` | UAV state (idle, moving, locked) | encodes movement and detection behavior |
| `zone` | spatial category (e.g., hotspot / normal) | affects local scoring rule |
| `uncertainty` | exploration incentive | decreases with visits, encourages exploration |
| `visit_penalty` | revisit discouragement | penalizes repeated visits to same cell |
| `shared_penalty` | coordination discouragement | reduces clustering among nearby UAVs |

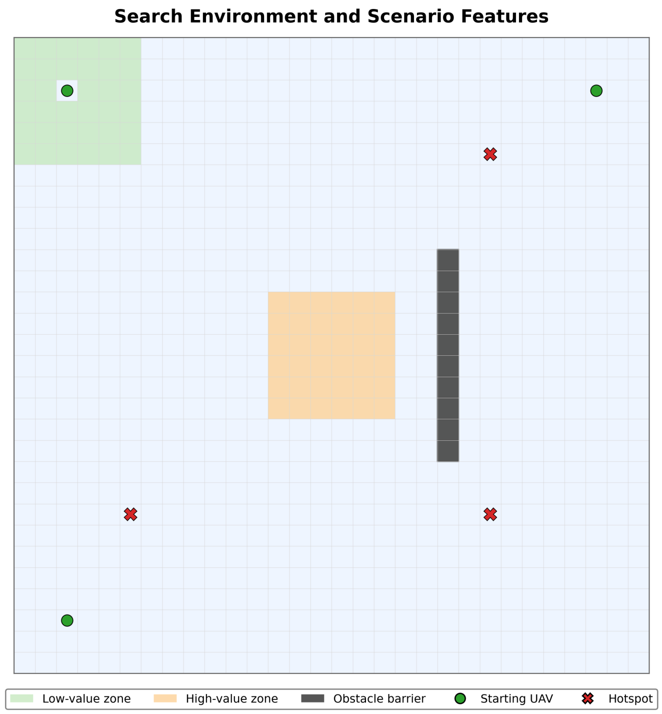


Fig. 1. Representative search environment including UAV starting positions, hotspot locations, obstacles, and spatially differentiated regions.

## E. Neighborhood

The model uses a Moore neighborhood of range 1. Therefore, each cell can interact with the eight surrounding cells. This supports both diffusion and local movement decisions. From a DEVS perspective, each grid cell represents an atomic model, where the state includes the probability field and UAV status, and the transition functions govern movement and probability updates based on local neighborhood interactions.

## F. Search Behavior

The initial assignment-style search logic was based mainly on local probability. In the final model, the UAV movement rule is more advanced. Each active UAV evaluates neighboring cells using a score that combines:

- local probability,
- local uncertainty,
- movement cost,
- revisit penalty,
- shared penalty,
- zone preference.

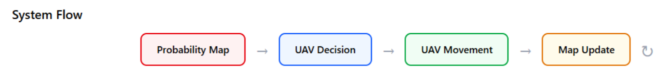


Fig. 2. High-level system flow: probability map evaluation, UAV decision, UAV movement, and map update.

This means the UAV is not simply chasing the highest probability value. Instead, it trades off exploitation of attractive regions against exploration of less-certain areas while avoiding repeated visits and excessive coordination overlap.

A conceptual form of the implemented local scoring rule is:

$$\text{Score}(n) = \text{zone\_bonus}(n)\,\frac{w_p\,p(n) + w_u\,u(n)}{d(n)+1} - w_v\,v(n) - w_s\,s(n) \quad (1)$$

where $p(n)$ is the probability of neighbor $n$, $u(n)$ is its uncertainty, $d(n)$ is the travel distance, $v(n)$ is the revisit penalty, and $s(n)$ is the shared penalty. This formulation reflects the implementation more closely than a purely additive score because the zone effect acts as a multiplier and the probability-uncertainty term is distance-weighted.

## G. System Flow

At a high level, the system evolves through a repeated loop:

1) evaluate the current probability map and local state,
2) compute UAV decisions using the local score rule,
3) move UAVs across the grid,
4) update the probability map, uncertainty, and penalties.

## H. Coordination Strategies

The project compares several coordination strategies, each defined operationally rather than only conceptually.

- **Single UAV baseline**: a reference case with only one UAV searching the environment.
- **Independent UAVs**: multiple UAVs use the same local movement rule but do not apply shared-information penalties. As a result, they may converge toward the same attractive regions and generate overlap naturally.
- **Partitioned deployment**: UAVs are initialized in separated starting regions, which induces spatial separation and reduces early overlap. However, this is not a strict hard partition of the grid; later movement is still governed by the local scoring rule.
- **Shared information**: UAVs are coordinated indirectly through penalty-based local information. In this case,

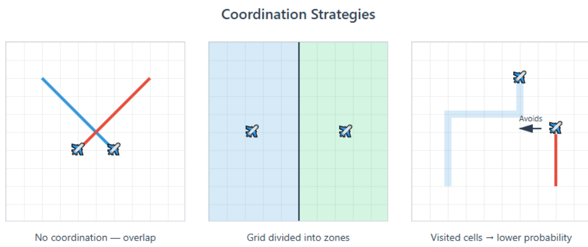


Fig. 3. Conceptual comparison of coordination strategies in the multi-UAV search model.

shared penalties discourage clustering and repeated exploration, but no explicit message-passing or centralized communication protocol is used.

- **Diffusion sensitivity**: the coordination rule is kept fixed while the diffusion parameter $\alpha$ is changed in order to study how field smoothness affects search behavior.

A conceptual comparison is shown in Fig. 3. These definitions are important because they clarify that the strategies differ in operational mechanism, not just in interpretation.

### *I. Probability Update*

The base probability update is diffusion-like:

$$p_{t+1} = (1 - 8\alpha)p_t + \alpha \sum_{n \in N} p_n$$

where $N$ denotes the Moore neighborhood.

This equation expresses a local redistribution of probability from neighboring cells. A low value of $\alpha$ preserves stronger local gradients, while a high value smooths the field more rapidly.

In the final model, probability update is also affected by:

- **UAV visit reduction**: visiting a cell reduces its local attractiveness;
- **shared-information reduction**: nearby UAV presence can further reduce probability;
- **pinned hotspots**: hotspot probability is restored to 1.0;
- **zone-aware diffusion**: different zones can use different diffusion intensities;
- **obstacle blocking**: obstacles prevent occupancy and suppress diffusion contribution.

### *J. Formal Cell-DEVS Specification*

The model is defined as a Cell-DEVS system over a two-dimensional non-wrapped grid with shape $(30, 30)$, origin $(0, 0)$, Moore neighborhood of range 1, transport delay, and output delay of 1 simulation time unit.

To complement the conceptual description, each cell is represented formally as a DEVS atomic component:

$$M = \langle X, S, Y, \delta_{int}, \delta_{ext}, \lambda, ta \rangle \tag{2}$$

where:

- $X$ is the set of input events, corresponding to neighboring cell states;
- $S$ is the set of cell states, defined as

  $S = \{\text{prob}, \text{uav}, \text{zone}, \text{uncertainty}, \text{visit\_penalty}, \text{shared\_penalty}\};$
- $Y$ is the set of output events, representing updated cell states communicated to neighbors;
- $\delta_{int}$ is the internal transition function, which updates the local cell state;
- $\delta_{ext}$ is the external transition function, which incorporates the influence of neighboring cells;
- $\lambda$ is the output function, which emits the updated state after each transition;
- $ta$ is the time advance function, defined here as a constant value of 1 simulation time unit.

This formal specification captures the key structure of the model while remaining consistent with the implementation: each cell updates through local interaction, probability evolution, and UAV-state transitions driven only by neighborhood information.

### *K. UAV Transition Rules*

At a conceptual level:

1) an active UAV evaluates neighboring cells using the local score;
2) the best-scoring direction is selected;
3) if no valid move exists or the current cell remains strongly attractive, the UAV can remain or lock;
4) empty cells detect incoming directional movement from neighbors;
5) directional movement states clear after movement is executed.

### *L. Lock State Interpretation*

A UAV may enter a lock state when the current cell remains sufficiently attractive under the local decision rule. In practice, this represents a high-confidence local detection or hold condition. However, lock does not necessarily imply a perfect one-to-one correspondence between a UAV and an exact hotspot cell. Instead, hotspot detection is interpreted at the system level through the emergence of stable high-confidence occupancy patterns. This interpretation is consistent with the simulation videos and with the way detection is measured in the post-processing scripts.

### *M. Uncertainty and Penalties*

The model introduces additional variables that were not present in the earlier assignment-style version:

- **uncertainty** decreases when a cell is visited and otherwise decays,
- **visit penalty** increases when a cell is revisited,
- **shared penalty** increases under shared-information coordination.

These variables provide memory-like effects at the cell level, even if the UAV itself does not carry a separate global memory map.

### N. Implementation in Cadmium

The project is implemented in C++ using Cadmium Cell-DEVS. The main implementation files are:

- `uav_search_state.hpp`: defines the extended cell state,
- `uav_search_cell.hpp`: implements the local update rule,
- `main.cpp`: builds and runs the grid model.

The project also includes a reproducible experiment pipeline:

- `build_sim.sh`: builds the project,
- `check.sh`: validates the binary, configs, and test execution,
- `run.sh`: runs all experiments and saves logs,
- `metrics.py`: computes reported metrics,
- `visualize.py`: produces figures for the report.

This aligns with the course requirement that the code should run cleanly, include scripts, and be easy to reproduce.

Taken together, the multi-agent extension, enriched cell state, local coordination penalties, obstacle-aware and zone-aware dynamics, and the inclusion of robustness experiments with a reproducible pipeline make this an advanced Cell-DEVS extension of the original Assignment 2 model.

## IV. Simulation Results

### A. Experimental Setup

The following experiments were executed:

- **Exp0**: single-UAV baseline on a $30 \times 30$ grid with one UAV and one hotspot.
- **Exp1**: three independent UAVs on a $30 \times 30$ grid with three hotspots.
- **Exp2**: partitioned-start deployment with three UAVs on a $30 \times 30$ grid with three hotspots.
- **Exp3**: shared-information coordination with three UAVs on a $30 \times 30$ grid with three hotspots.
- **Exp4a**: low-diffusion scenario on a $30 \times 30$ grid with three UAVs and three hotspots.
- **Exp4b**: high-diffusion scenario on a $30 \times 30$ grid with three UAVs and three hotspots.
- **Exp5a**: independent stress scenario on a $50 \times 50$ grid with four UAVs and five hotspots.
- **Exp5b**: partitioned stress scenario on a $50 \times 50$ grid with four UAVs and five hotspots.
- **Exp5c**: shared-information stress scenario on a $50 \times 50$ grid with four UAVs and five hotspots.

The first group of experiments (Exp0–Exp4b) was designed to evaluate the baseline coordination strategies and the effect of diffusion in a controlled $30 \times 30$ environment. These experiments provide the main reference for comparing single-UAV and multi-UAV search behavior under different coordination rules.

The second group of experiments (Exp5a–Exp5c) extends the evaluation to a more complex setting. In this case, the grid size is increased from $30 \times 30$ to $50 \times 50$, the number of UAVs is increased from three to four, and the number of hotspot locations is increased from three to five. These experiments were included to test whether the main coordination patterns observed in the base scenarios remain valid under increased spatial size, more hotspot locations, and greater coordination burden.

All experiments include obstacle constraints, and each scenario was run for 1000 simulation steps. The comparison is therefore controlled within each scenario family: only the coordination strategy or diffusion setting is changed, while the rest of the environment configuration remains fixed. This makes it possible to attribute the observed differences in behavior directly to the coordination logic and probability-field dynamics rather than to unrelated changes in the simulation setup.

### B. Stress-Scenario Extension

To further evaluate the robustness of the proposed coordination strategies, an additional set of stress scenarios was tested. Unlike the baseline and diffusion experiments on a $30 \times 30$ grid, these extended cases use a larger $50 \times 50$ environment with four UAVs, five hotspots, and obstacle constraints. Three coordination variants were evaluated in this setting: independent search (Exp5a), partitioned deployment (Exp5b), and shared-information coordination (Exp5c).

These stress scenarios were designed to test whether the coordination patterns observed in the base experiments remain valid under increased spatial size, more targets, and greater coordination burden. In this sense, Exp5 serves as a robustness- and scalability-oriented extension of the main experiment set rather than a replacement for the original comparisons.

### C. Evaluation Metrics

The experiments are evaluated with:

- **Coverage**: percentage of unique grid cells visited;
- **Overlap**: percentage of visited cells that were revisited;
- **Revisit intensity**: extra visits relative to total visits;
- **First detection time**: first timestep at which a hotspot is effectively found;
- **Hotspots found**: number of detected hotspots.

In post-processing, a hotspot is counted as found when the simulation output indicates a stable high-confidence detection associated with that hotspot location; this metric is therefore derived from the logged system-level detection behavior rather than from a strict one-to-one lock-to-hotspot correspondence.

These metrics were chosen because they jointly reflect exploration breadth, redundancy, speed, and detection success. Together, they provide a more complete picture of search performance than any single metric alone. For example, a strategy may detect targets quickly but still perform poorly in terms of redundancy, or it may achieve high coverage but at the cost of excessive repeated visits. In addition to the static figures presented in this report, dynamic visualizations of the simulation were generated from the output logs. These videos illustrate the temporal evolution of UAV movement, lock behavior, and probability-field diffusion. They are especially useful for interpreting coordination patterns that may be harder to observe from final-state images alone. The videos therefore complement the quantitative and qualitative analysis by providing a direct view of the spatiotemporal dynamics of the search process.

### D. Qualitative Results

Figure 4 shows the final probability fields and UAV states at the end of the simulation for the six base experiments.

Figure 4 provides a qualitative overview of how each coordination strategy shapes the final search pattern. The single-UAV baseline explores more slowly and detects fewer hotspots, which is reflected in its more limited spatial spread. In contrast, the multi-UAV cases cover the environment more aggressively and are able to detect all three hotspots, although the final field structure differs significantly across strategies.

Several important visual patterns can be observed. First, the obstacle barrier remains clearly visible as a persistent low-probability vertical structure, confirming that movement and diffusion constraints continue to influence the search throughout the run. Second, the shared-information and partitioned strategies produce visibly different spatial organizations, even though both aim to reduce redundant exploration. Third, the diffusion-sensitivity cases reveal how strongly $\alpha$ affects the shape of the field: lower diffusion preserves sharper local gradients around hotspot locations, whereas higher diffusion smooths the field and reduces local contrast. Thus, the qualitative results already suggest that coordination changes not only detection speed, but also the geometry of the search process itself.

### E. Stress-Scenario Qualitative Results

Figure 5 shows the final probability fields for the three Exp5 stress-scenario variants.

Figure 5 shows that the qualitative structure of the search process remains consistent in the larger scenario. All three strategies detect four out of five hotspots, while the obstacle structure continues to shape the field and constrain movement. The partitioned case preserves clearer spatial separation and slightly improves coverage, whereas the shared-information case produces localized suppression regions around hotspot areas. These results suggest that the coordination mechanisms remain meaningful in a larger and more demanding environment, although the overall coverage remains limited relative to the full grid size.

### F. Coverage Over Time

Figure 6 shows the growth of coverage over simulation time for the six base experiments.

Figure 6 captures both the speed of exploration and the final explored portion of the grid. The single-UAV baseline increases coverage slowly, which confirms the limitations of individual-agent search in a relatively large environment. By contrast, all multi-UAV strategies reach their final coverage much earlier, showing the advantage of parallel exploration.

However, the final coverage levels differ substantially across the multi-UAV cases. The low-diffusion experiment achieves the highest final coverage, while the partitioned and high-diffusion cases plateau at lower values. This indicates that simply spreading UAVs apart at the start is not enough to guarantee broad long-term exploration. Later behavior is still strongly shaped by the evolving probability field and by the local decision rule. It is also worth noting that the final coverage values remain relatively small compared with the full $30 \times 30$ grid. This is because the UAVs are not performing exhaustive uniform coverage; instead, they are being guided by attractive regions in the probability field.

### G. Stress-Scenario Coverage Over Time

Figure 7 compares coverage growth over time for the three stress-scenario variants.

In the stress scenario, all three strategies reach their final coverage very quickly and then plateau for the remainder of the simulation. The partitioned variant achieves the highest final coverage, while the independent and shared-information cases remain slightly lower and nearly identical. This result is consistent with the earlier observation that partitioning is effective at reducing redundant early exploration, although the final differences remain small in absolute terms. Because the search space is larger in Exp5, the low final coverage values also highlight the difficulty of achieving broad exploration under locally guided search.

### H. Independent vs Shared Information

Figure 8 compares the temporal evolution of the independent and shared-information scenarios.

At early times, both strategies react strongly to hotspot locations because local probability gradients dominate movement decisions. As the simulation progresses, the shared-information case alters the search pattern more clearly by reducing the attractiveness of areas that are already being explored or influenced by nearby UAV activity. This leads to a visibly different spatial evolution compared with the independent case.

The figure shows that shared-information coordination changes where UAVs concentrate and tends to spread search pressure more effectively across the grid. However, it also shows that shared information does not fully eliminate clustering. Strong hotspot-driven gradients still attract multiple UAVs toward similar areas. Therefore, shared information improves search behavior without guaranteeing zero overlap. This observation is consistent with the quantitative results, where the shared-information case achieves improved coverage relative to the independent case but still exhibits redundancy.

### I. Diffusion Sensitivity

Figure 9 compares the low- and high-diffusion experiments.

Figure 9 illustrates how the diffusion parameter affects the evolution of the probability field. In the low-diffusion case, the field remains more localized for a longer time, which preserves stronger gradients around hotspot sources. These

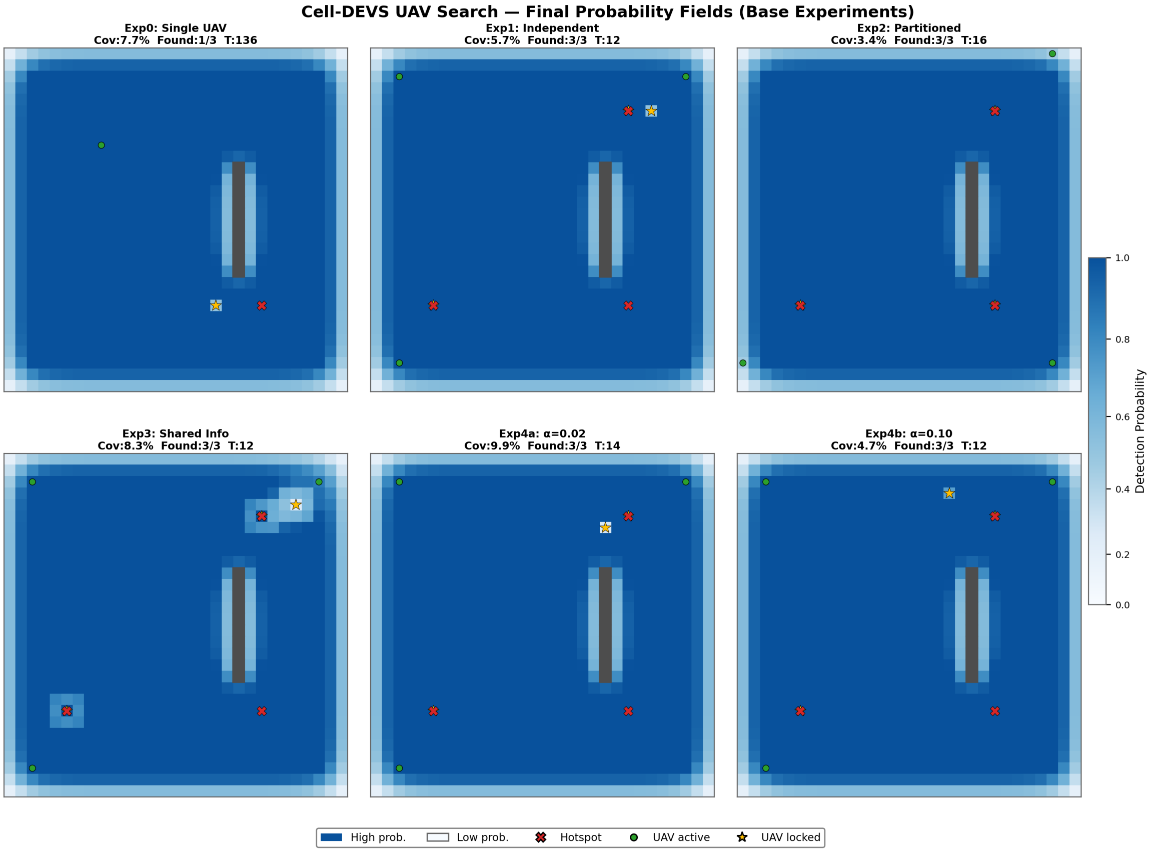


Fig. 4. Final probability fields and UAV states at the end of the simulation for the six base experiments.

sharper gradients provide clearer local guidance to the UAVs and support higher final coverage. However, they also increase the likelihood that multiple UAVs repeatedly return to the same attractive regions.

In the high-diffusion case, the field smooths more rapidly, reducing local contrast and weakening the directional guidance provided by hotspots. As a result, the search becomes less strongly structured by high-probability regions. This helps explain why the high-diffusion configuration produces lower final coverage. The comparison highlights that diffusion is not just a background environmental parameter; it directly changes how informative the field is for decentralized search decisions.

### J. Quantitative Metrics

Figure 10 summarizes the main quantitative metrics across the six base experiments.

Figure 10 is one of the most important result summaries because it condenses the main trade-offs into a single view. The single-UAV baseline performs worst in terms of detection speed and hotspot count, confirming that one UAV is insufficient for efficient search over the full grid. All multi-UAV strategies substantially reduce first detection time and detect all three hotspots, showing the strong benefit of multi-agent search.

At the same time, the figure shows that better detection performance does not automatically imply better overall efficiency. The partitioned strategy achieves the lowest overlap, indicating the strongest reduction in redundant search, but it also produces relatively low final coverage. The low-diffusion case achieves the highest coverage, but it also exhibits the highest overlap, showing that broader exploration may come at the cost of repeated attention to the same attractive areas. Therefore, the results make clear that maximizing coverage and minimizing redundancy are competing objectives rather than identical goals.

Figure 11 provides a normalized multi-dimensional comparison of the strategies.

Unlike the bar-chart metrics, Fig. 11 does not show raw values directly. Instead, it visualizes the relative performance of each experiment across normalized coverage, detection, efficiency, and speed. This makes it easier to compare the overall shape of each strategy's strengths and weaknesses. For example, the low-diffusion case expands strongly in coverage but contracts in efficiency because of its higher overlap, while the partitioned case shows the opposite pattern. The radar chart is therefore useful as a compact summary, but it should be interpreted together with the raw values in Fig. 10 and Table II.

Table II reports the final quantitative metrics obtained from the full simulation runs executed using `run.sh`. These results are used for all subsequent analysis.

Several observations follow from Table II. First, all multi-

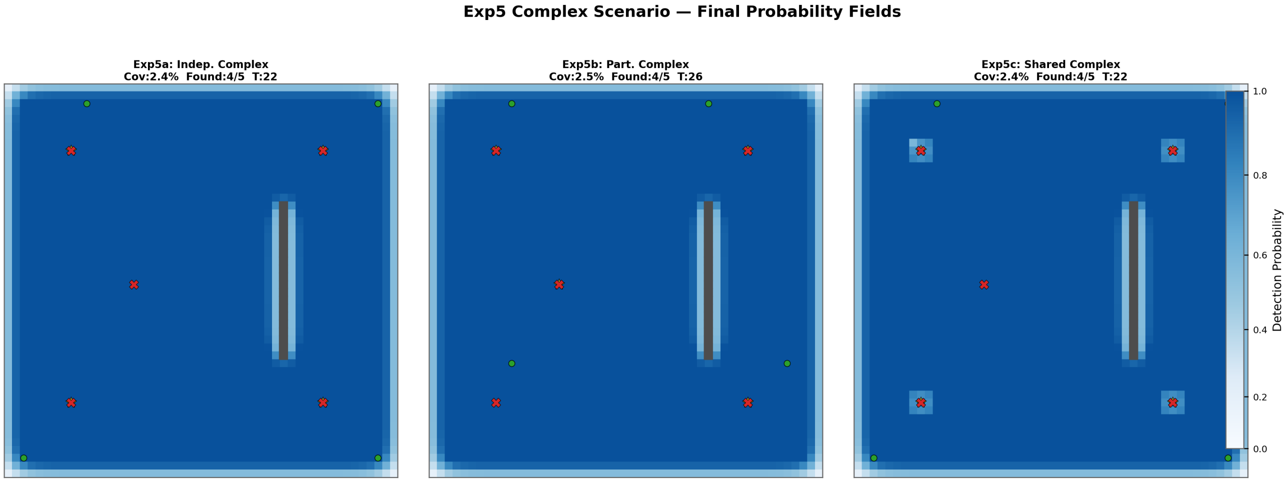


Fig. 5. Final probability fields for the $50 \times 50$ stress scenario with four UAVs and five hotspot locations: independent search (Exp5a), partitioned deployment (Exp5b), and shared-information coordination (Exp5c).

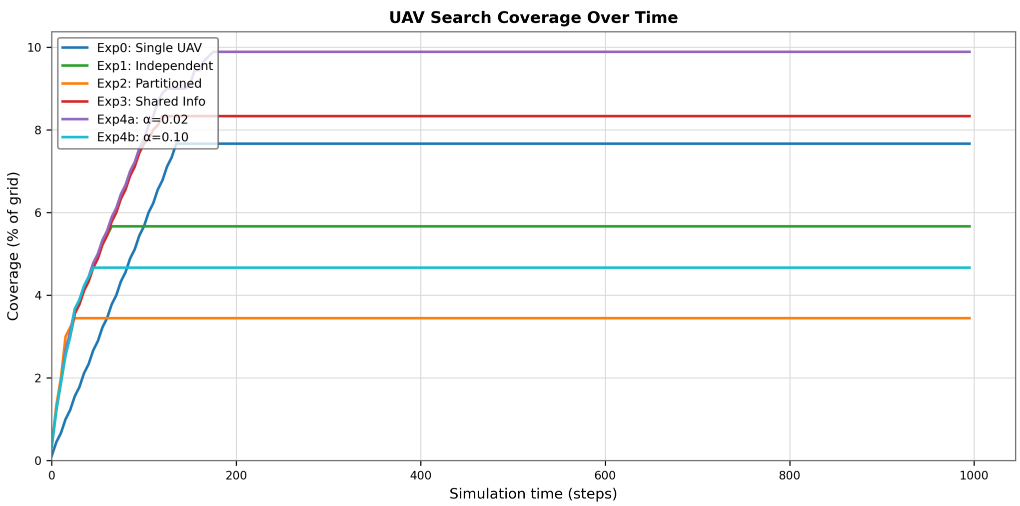


Fig. 6. Coverage growth over simulation time for the six base experiments.

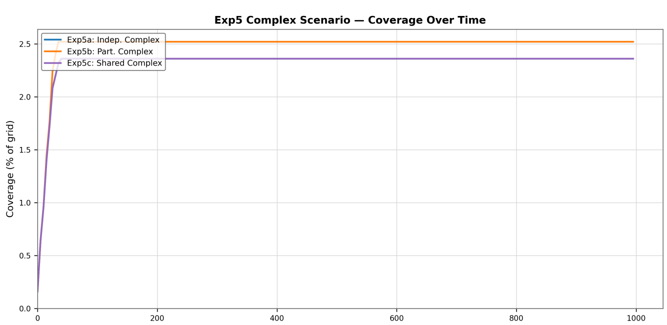


Fig. 7. Coverage over time for the Exp5 stress-scenario variants.

TABLE II

SUMMARY OF EXPERIMENTAL RESULTS FOR ALL COORDINATION STRATEGIES OVER THE FULL SIMULATION RUN.

| Experiment | Cov. (%) | Overlap (%) | Intensity (%) | $T_{\text{first}}$ | Found |
|---|---|---|---|---|---|
| Exp0 Single UAV | 7.7 | 0.0 | 0.0 | 136 | 1/1 |
| Exp1 Independent | 5.7 | 3.9 | 3.8 | 12 | 3/3 |
| Exp2 Partitioned | 3.4 | 0.0 | 0.0 | 16 | 3/3 |
| Exp3 Shared Info | 8.3 | 8.0 | 7.4 | 12 | 3/3 |
| Exp4a $\alpha = 0.02$ | 9.9 | 23.6 | 19.1 | 14 | 3/3 |
| Exp4b $\alpha = 0.10$ | 4.7 | 4.8 | 4.5 | 12 | 3/3 |
| Exp5a | 2.4 | 1.7 | 1.7 | 22 | 4/5 |
| Exp5b | 2.5 | 0.0 | 0.0 | 26 | 4/5 |
| Exp5c | 2.4 | 1.7 | 1.7 | 22 | 4/5 |

UAV cases detect all three hotspots much faster than the single-UAV baseline. Second, Exp4a achieves the highest coverage, which confirms the qualitative impression that sharper local gradients can improve exploratory spread. Third, Exp2 achieves the lowest overlap among the multi-UAV strategies, showing that simple spatial separation can effectively reduce redundancy. Finally, the table reinforces an important design

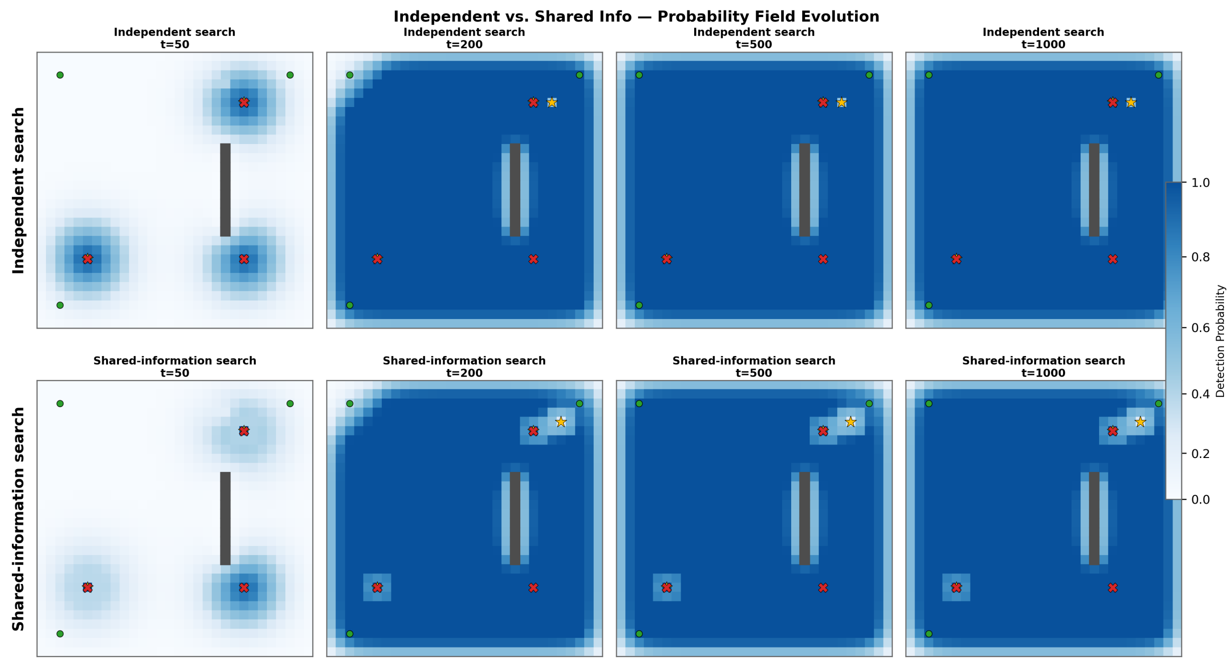


Fig. 8. Probability-field evolution for independent search and shared-information search.

trade-off: the strategy with the highest coverage is not the one with the lowest overlap.

### *K. Stress-Scenario Quantitative Metrics*

Figure 12 summarizes the main quantitative metrics for the Exp5 stress-scenario variants.

Figure 12 shows that all three stress-scenario variants detect four out of five hotspots, but they differ slightly in coverage, overlap, and first detection time. The partitioned variant achieves the highest coverage and zero overlap, but it also has the slowest first detection time. By contrast, the independent and shared-information cases detect hotspots earlier, but both exhibit a small amount of overlap. These results reinforce the same trade-off observed in the main experiments: stronger deconfliction tends to reduce redundancy, but may also reduce responsiveness or adaptability during search.

### *L. Uncertainty Evolution*

Figure 13 shows the evolution of the average grid uncertainty over time for the six base experiments.

Uncertainty provides a global measure of how much ambiguity remains in the probability field during the search process. This metric is useful because it summarizes how informative or uncertain the overall environment remains as the simulation progresses. At early times, uncertainty increases as diffusion spreads information away from the hotspot sources and produces a broader range of intermediate probability values across the grid. Later, uncertainty decreases as UAV visits, locked detections, and local suppression effects make the field more structured and reduce ambiguity.

In this sense, Fig. 13 complements the coverage and detection metrics by showing not only where the UAVs moved, but also how quickly each strategy reduced uncertainty over the search space. The uncertainty curves therefore reveal another dimension of coordination quality: two strategies may have similar detection times, yet differ in how efficiently they organize information and reduce ambiguity across the environment.

### *M. Stress-Scenario Uncertainty Evolution*

Figure 14 shows the average grid uncertainty over time for the three stress-scenario variants.

The uncertainty curves in Fig. 14 follow the same general pattern observed in the earlier experiments: uncertainty rises initially as probability diffuses through the larger grid, then decreases as the search process becomes more structured. The shared-information case retains slightly higher uncertainty for longer, while the partitioned case drops more quickly after the early peak. This suggests that even in the larger environment, the choice of coordination policy affects not only search trajectories but also the rate at which ambiguity is reduced across the grid.

### *N. Main Findings*

The main findings of the experiments can be summarized as follows:

- Multi-UAV search dramatically improves hotspot-detection speed compared with the single-UAV baseline.
- Partitioned deployment minimizes overlap, but this comes at the cost of reduced total coverage.
- Shared-information coordination improves coverage relative to independent search, but it does not eliminate redundant convergence.

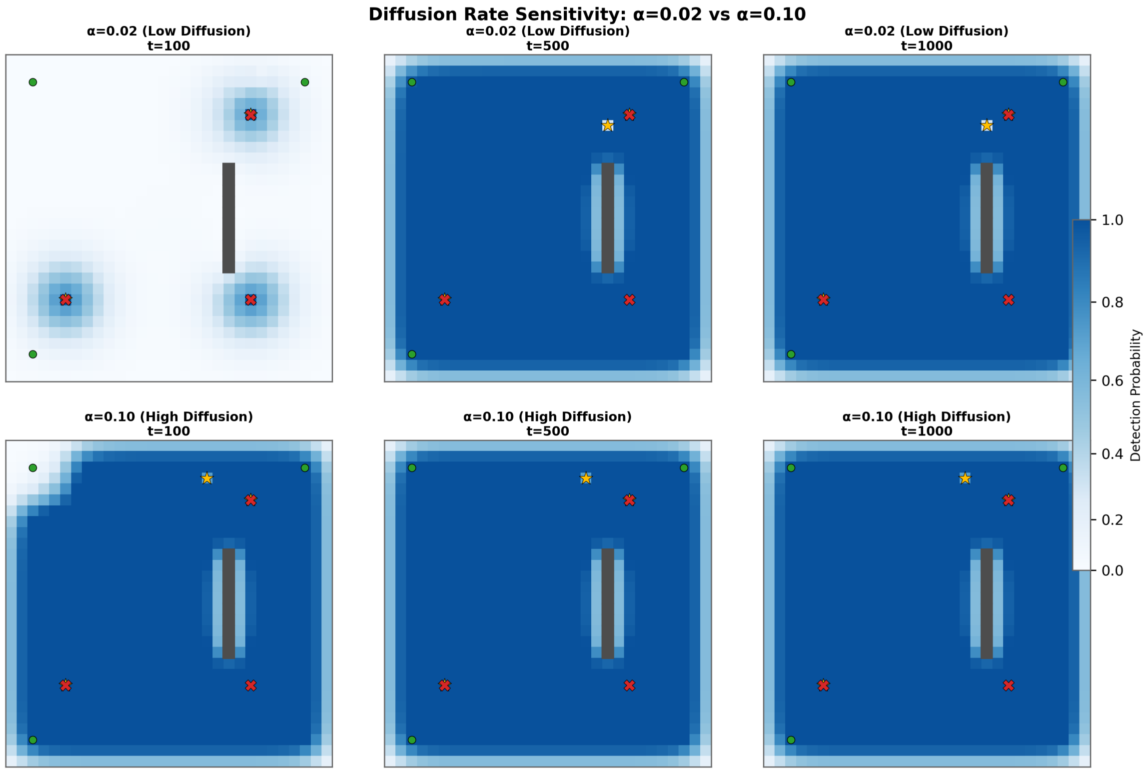


Fig. 9. Effect of diffusion rate on the probability field at selected times.

- Low diffusion increases exploration breadth by preserving sharper gradients, but it also raises redundancy.
- High diffusion smooths the field more rapidly and reduces final coverage by weakening local directional guidance.

These findings show that no single strategy is uniformly best across all metrics. Instead, the results reflect a trade-off between speed, coverage, and redundancy, which is typical of decentralized multi-agent search systems.

### *O. Discussion*

The results obtained in this study align with the general understanding in multi-UAV coordination that increasing the number of agents significantly improves detection speed, but also introduces coordination challenges such as redundancy and overlap [4]. In this work, these trade-offs are explicitly captured and analyzed within a formal Cell-DEVS modeling framework, allowing for a structured comparison of different coordination strategies.

The single-UAV baseline highlights the fundamental limitations of individual-agent search. Exploration progresses slowly, and only one hotspot is detected within the simulation horizon. This scenario establishes a clear reference point, demonstrating the necessity of multi-agent systems for large-scale search tasks.

When multiple UAVs are introduced, detection performance improves dramatically across all strategies, with most configurations successfully identifying all hotspots. However, the manner in which UAVs coordinate their movement has a substantial impact on overall efficiency. The independent strategy performs reasonably well, achieving fast detection and moderate coverage, but it does not incorporate any mechanism to reduce redundancy. As a result, UAVs tend to revisit similar regions, leading to inefficient use of search effort.

The partitioned-start strategy demonstrates that even simple spatial coordination can significantly influence system behavior. By distributing UAVs across different regions of the grid at initialization, early overlap is reduced and search trajectories become more distinct. This results in the lowest overlap among all strategies. However, this improvement comes at the cost of reduced global coverage, suggesting that strict spatial separation may limit the ability of UAVs to adapt to dynamically evolving probability fields.

The shared-information strategy introduces a more dynamic form of coordination by modifying the probability field based on neighboring UAV activity. This discourages clustering and promotes dispersion, leading to improved coverage compared to the independent case. Nevertheless, redundancy is not fully eliminated. UAVs are still indirectly attracted to high-probability regions, which remain strong drivers of movement. This indicates that while local information sharing can reshape exploration patterns, it is insufficient on its own to fully prevent overlap.

The influence of the diffusion parameter further illustrates the sensitivity of the system to environmental dynamics. In the low-diffusion case, the probability field retains sharper

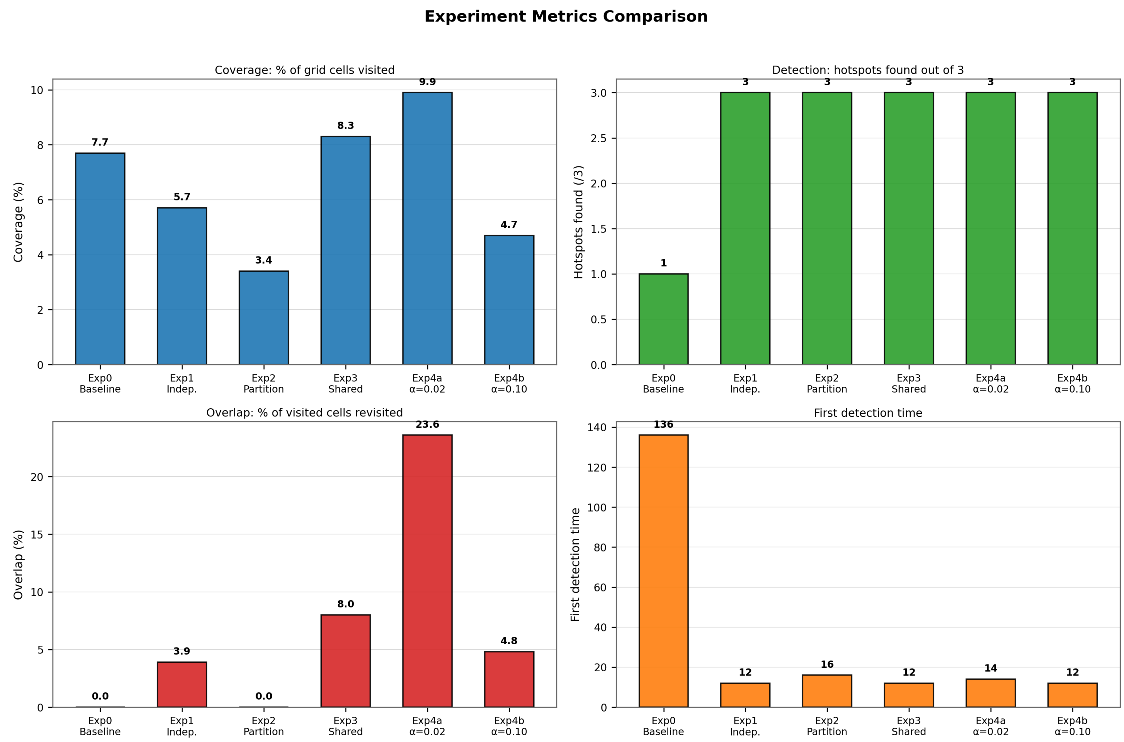


Fig. 10. Quantitative metrics comparison for the six base experiments.

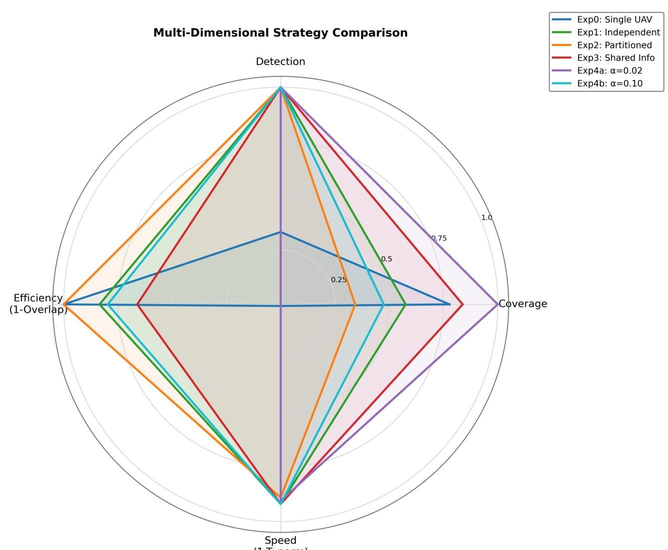


Fig. 11. Radar summary comparing normalized coverage, detection, efficiency, and speed across the six base experiments.

gradients for a longer period, making high-value regions more distinguishable. This enhances exploratory behavior and results in the highest overall coverage. However, it also increases revisit concentration, as UAVs are repeatedly drawn to the same attractive regions. In contrast, high diffusion smooths the probability field more rapidly, reducing the strength of local gradients. This leads to a more uniform exploration pattern with slightly lower redundancy, but also reduces the effectiveness of directed search, resulting in lower coverage.

Taken together, these observations reveal that the core design challenge in multi-UAV search is not simply maximizing coverage or minimizing overlap independently. Instead, the objective is to achieve a balanced coordination strategy that simultaneously satisfies several competing goals:

- rapid and reliable hotspot detection,
- efficient coverage of the search space,
- minimization of redundant visits,
- robust operation under decentralized and locally available information.

The results clearly demonstrate that different coordination mechanisms emphasize different aspects of this balance. Strategies that promote exploration, such as shared information and low diffusion, tend to increase overall coverage but may also lead to higher redundancy due to repeated attraction toward high-probability regions. On the other hand, strategies that explicitly reduce overlap, such as partitioned deployment, can limit redundancy but may constrain the system's ability to explore globally.

These findings highlight a fundamental trade-off in decentralized multi-agent systems: exploration efficiency and redundancy control are inherently competing objectives. Achieving optimal performance therefore requires adaptive coordination mechanisms that can dynamically adjust behavior based on the current state of the environment. Rather than relying on fixed strategies, future approaches may benefit from integrating

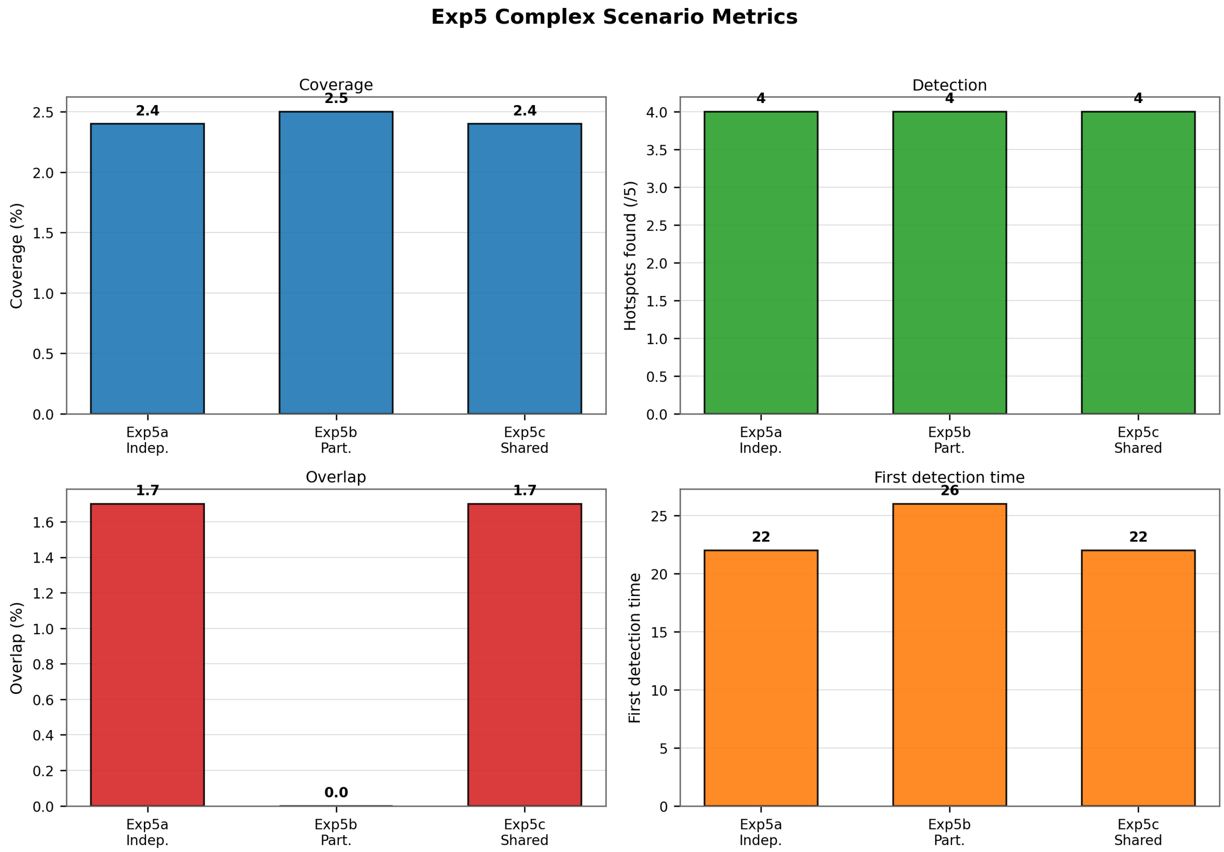


Fig. 12. Quantitative metrics for the Exp5 stress-scenario variants.

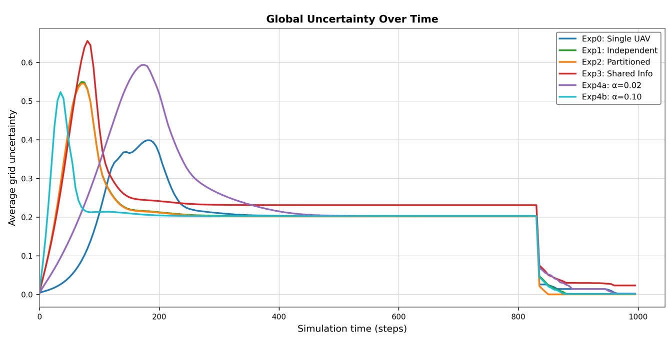


Fig. 13. Average grid uncertainty over time for the six base experiments.

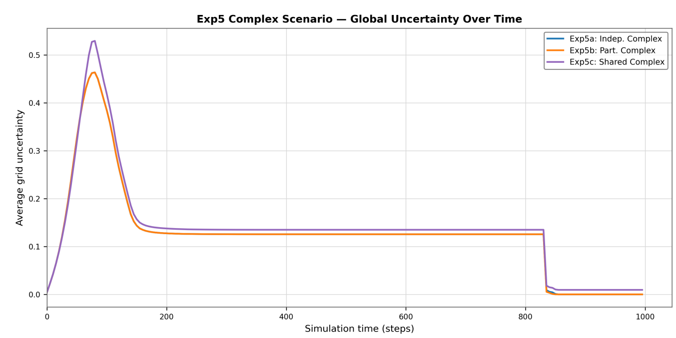


Fig. 14. Average grid uncertainty over time for the Exp5 stress-scenario variants.

learning-based or context-aware policies that continuously balance exploration and exploitation.

## V. Conclusions

This project developed an advanced Cell-DEVS model for multi-UAV search and coordination using Cadmium. The proposed model extends earlier work by incorporating multiple UAVs, enriched state variables, uncertainty-aware decision-making, coordination penalties, and configurable spatial features.

The experimental results demonstrate that coordination strategy has a significant impact on system performance. Multi-UAV configurations substantially improve hotspot detection speed compared to a single UAV, but redundancy remains a key challenge. Partitioned deployment reduces overlap, shared-information strategies reshape exploration patterns without fully eliminating redundancy, and the diffusion parameter strongly influences the evolution and informativeness of the probability field.

In addition, the stress-scenario experiments on a larger $50\times$

50 grid with four UAVs and five hotspots showed that the main coordination trade-offs observed in the base experiments remain present under increased spatial size, more targets, and greater coordination burden.

Overall, the project satisfies the objective of designing an improved Cell-DEVS model with extended rules and multiple experimental scenarios, as required for the course term project. More importantly, it demonstrates that Cell-DEVS provides a rigorous and flexible framework for analyzing complex spatial coordination problems in decentralized multi-agent systems.

The results highlight the effectiveness of local, decentralized coordination mechanisms in shaping system-level behavior, and emphasize the importance of combining formal modeling with experimental validation. While the model uses a regular grid topology, it introduces effective asymmetry through spatial variations and transition rules, extending the expressive power of the base Cell-DEVS formulation.

### A. Limitations

While the proposed model demonstrates strong performance and flexibility, several limitations remain. First, UAV decision-making is fully local, with no explicit communication between agents. This limits the ability to coordinate globally and may contribute to residual overlap in explored regions. Second, the environment is static and simplified, with predefined zones and obstacles. Real-world scenarios would include dynamic targets, uncertain terrain, and sensor noise. Finally, the model assumes fixed parameters such as diffusion rate and penalty weights, which may not be optimal across all scenarios. Adaptive or learning-based parameter tuning could further improve performance.

### B. Future Work

Several directions can extend this work toward more realistic and intelligent UAV coordination. A key improvement is the integration of explicit communication between UAVs, enabling global coordination strategies beyond local interactions. This could reduce redundancy and improve coverage efficiency in large-scale environments.

Another promising direction is the incorporation of learning-based approaches. Reinforcement learning or adaptive control could allow UAVs to dynamically adjust parameters such as diffusion rate, movement policy, and revisit penalties based on the environment and mission goals.

The current model assumes a static environment. Future work should consider dynamic scenarios, including moving targets, time-varying probability fields, and environmental uncertainty. Incorporating sensor noise and partial observability would further enhance realism.

Finally, extending the model to larger grids and higher UAV densities would allow scalability analysis. Real-time implementation and integration with physical UAV systems or hardware-in-the-loop simulation would bridge the gap between simulation and practical deployment.

## References


[1] B. P. Zeigler, H. Praehofer, and T. G. Kim, *Theory of Modeling and Simulation*, 2nd ed. San Diego, CA, USA: Academic Press, 2000.

[2] G. Wainer, *Discrete-Event Modeling and Simulation: A Practitioner's Approach*. Boca Raton, FL, USA: CRC Press, 2009.

[3] L. Moshagen, C. Castelar Wembers, and G. Schildbach, "Probability maps and search strategies for automated UAV search in the Wadden Sea," in *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2022, pp. 10969–10975.

[4] A. Aminzadeh and A. M. Khoshnood, "Multi-UAV cooperative search and coverage control in post-disaster assessment," *IEEE Access*, vol. 9, pp. 117045–117058, 2021.